AdoptSense: Scaling a Two-Sided Pet Adoption Platform

Video Pitch: OneDrive[1,2] | **Repo:** Github[3] | **Deployed App:** Streamlit[4]| **MVP Demo Video:** OneDrive[1,5]| **Words:** 2749

1. Introduction

AdoptSense is a non-profit, AI-powered pet adoption platform that connects animal shelters with prospective adopters. The working prototype (Minimum Viable Product, or MVP) combines three things in one tool: an AI model that predicts how quickly a pet listing will be adopted, a recommendation engine that tells shelter managers what to change to improve their listings, and a two-sided marketplace where households browse, save, and contact shelters directly. This memo asks the CEO to approve the move from MVP to a 12-month commercial pilot in Portugal and a first-year budget of roughly €30,000 that lets the volunteer team transition into a sustainable non-profit operation.

The case is concrete. In Portugal alone, 42,718 animals entered municipal shelters in 2024, 10,171 were never adopted, and 2,421 were euthanized. Across Europe, the pet market is worth €6.74 billion and reaches 139 million households (FEDIAF, 2024); a 2024 German Animal Welfare Federation survey found 69 percent of German shelters at capacity. The workflow producing these numbers is uniform: a pet arrives, a listing goes up in a hurry, the listing sits, and a manager intervenes only weeks later when the kennel is full. By then, boarding costs have piled up and the animal has become harder to place.

AdoptSense closes the information gap that drives this. Our prediction engine tells a shelter which listings risk stagnating, and the recommendation layer specifies what to do: add a photo, lower the fee, rewrite the description, enhance the photo. The marketplace gives adopters a transparent, filterable place to find a pet and contact the shelter. The CEO has one decision to make: commit to a Portugal pilot and turn a working prototype into something that can grow; or don't.

We recommend watching the MVP Demo Video and test the deployed MVP since it explains the most important features and functionalities behind the relative complex solution.

2. Goals

We propose six measurable year-one goals (May 2026 – May 2027), paired so that throughput cannot be improved at the expense of welfare.

On throughput: reduce average length-of-stay in pilot shelters by 20 percent against each shelter's pre-AdoptSense baseline by May 2027; reduce the share of pets flagged as "High priority" (slowest class) by 5 percentage points against baseline; increase qualified household inquiries per active listing by at least 100

[1] https://novasbe365.sharepoint.com/sites/AIImpactonBusinessTXA11/_layouts/15/stream.aspx?id=%2Fsites%2FAIImpactonBusinessTXA11%2FDocumentos%20Partilhados%2FGeneral%2FRecording%20MVP%2FGroup11%5FTXA%5FYCPitch%2Emp4&ga=1&referrer=StreamWebApp%2EWeb&referrerScenario=AddressBarCopied%2Eview%2Ee03a95be%2Dbd54%2D48a5%2D8fe4%2Defba84df09d0

[2] For privacy reasons, only Nova SBE accounts can access the material on the Sharepoint. Hence please login to your Nova SBE account first.

[3] https://github.com/Collaborating00/AdoptSense

[4] https://adoptsense-test.streamlit.app

[5] https://novasbe365.sharepoint.com/sites/AIImpactonBusinessTXA11/_layouts/15/stream.aspx?id=%2Fsites%2FAIImpactonBusinessTXA11%2FDocumentos%20Partilhados%2FGeneral%2FRecording%20MVP%2FGroup11%5FTXA%5FVideo%5FMVP%2Emp4&ga=1&referrer=StreamWebApp%2EWeb&referrerScenario=AddressBarCopied%2Eview%2Ef031423c%2Da9ae%2D4a71%2Db51c%2D1a022ebd1cfa

percent versus pre-pilot baseline (internal MVP estimates suggest up to a 5x increase, which we treat as upside).

On adoption and growth: onboard 50 Portuguese shelters across Lisbon and Porto regions by May 2027, with a stretch path to 600 shelters across Southern Europe by year 3; reach 5,000 monthly active household accounts in Portugal.

On welfare and sustainability: keep the 12-month post-adoption reinsertion rate (our pet-flipping early warning) below 5 percent across all platform adoptions, and maintain a post-adoption Likert satisfaction score of at least 4.0 out of 5 from both shelters and households. If either welfare metric deteriorates by more than 10 percent quarter-on-quarter, listing-prioritization and featured-listing features pause automatically until the metrics recover.

3. Tenets

Every pet has a future, and AdoptSense's job is to help that future arrive faster. We measure success in animals who leave a kennel and find a home. Standard KPIs like user engagement, time on site, or the standard platform vanity metrics are not doing its justice. When trade-offs arise, animal welfare wins.

The purpose of AI is purely to give recommendations. No feature on AdoptSense will ever change a listing, set a fee, or contact an adopter without an explicit human click. Shelter staff and households are decision-makers, not executors of an algorithm they did not author.

Transparent over clever. Every prediction, every AI-rewritten description, every AI-enhanced photo is labelled and accompanied by an explanation a shelter manager can read and dispute. A recommendation a user cannot interrogate is a recommendation they will rightly ignore.

Adoption that lasts beats adoption that is fast. A successful adoption is a pet that stays in its new home, not one that returns three months later.

4. State of the Business

AdoptSense today is a working MVP and is feature-complete for pilot deployment. We launch in Portugal: over 200 municipal shelters, no incumbent intelligent platform, regulatory environment broadly aligned with EU AI Act expectations. Competition is fragmented (Petfinder, Adopt-a-Pet, Facebook Marketplace, individual shelter sites) — none combines prediction, recommendations, and a marketplace.

The MVP is a Python web application with three user roles: Shelter Manager, Private Household, and an Admin role for the AdoptSense team.

Financially, the model is a four-stream non-profit mix anchored on Software-as-a-Service (SaaS) shelter subscriptions. Pricing is adopter-friendly: free for households, free Basic tier for all shelters with unlimited listings and AI predictions, and a Pro tier at €49/month/shelter for the full KPI dashboard, listing prioritization, AI description improvement, and priority support. Phase 2 (year 1) targets 50 onboarded shelters, of which 20 convert to Pro, generating approximately €1,780/month against €2,200–3,000/month in operating costs — a break-even target. Phase 3 (year 2–3) scales to 600 shelters and ~€280,000 annual revenue at ~98 percent gross margin (Appendix A.4).

5. Lessons Learned

Four lessons from MVP development shape the pilot plan.

The model is trained on the wrong continent. Our classifier learned from PetFinder.my data - Malaysian pet listings - because no comparably large labelled European dataset exists publicly. Adoption preferences are locally specific: fur length, breed mix, age tolerance, and fee sensitivity vary by region, so the model will be systematically biased when used at inference in Portugal. The mitigation is built into the architecture: every new listing, adoption outcome, and operator decision feeds the next quarterly retraining cycle with explicit regional partitioning. From launch month one we collect that data; by month six we expect the first Portugal-stratified retrain.

Model performance is meaningful but not high in absolute terms. Accuracy of 40 percent means the model correctly predicts the adoption speed category in 4 out of 10 cases — twice the 20 percent a random guess yields on a five-class problem. The macro F1 score of 0.35 is a stricter accuracy measure that checks whether the model performs consistently across all five categories, not just the common ones; a perfect score is 1.0, random guessing lands around 0.20. The AUC of 0.75 on the "no adoption" class measures how confidently the model separates animals unlikely to ever be adopted from those that will — think of it as the model's ability to spot the hardest cases, which is the prediction that matters most operationally.

Is 0.35 high enough to launch? It depends on what the model is being asked to do. If AdoptSense were replacing shelter manager judgment, the answer would be no — but it is not. The model flags priority; the manager decides. The relevant comparison is not a perfect classifier, it is the status quo: no systematic prioritization at all in most Portuguese shelters.

Generative AI is the most loved feature and the most operationally risky. Shelter managers responded strongly to AI description rewriting and the Photo Studio; internal testing suggests AI-polished listings can be adopted noticeably faster (the company-presentation deck cites up to 46 percent faster, to be validated at pilot scale). But a hallucinated description that overstates a pet's health can produce a failed adoption. We learned to gate every AI-rewritten description through explicit shelter approval, keep the original always accessible, and label every AI-generated element with a visible disclosure badge.

A working prototype is not a working product. The MVP runs on Streamlit and SQLite. Migrating the UI layer to FastAPI, moving the database to PostgreSQL with cloud photo storage, and executing Data Processing Agreements (DPAs) with Google and Hugging Face are non-negotiable pre-launch tasks, scoped into the year-one budget.

6. Strategic Priorities

6.1 Proposed Solution and Project Scoping

The year-one priority is to convert the MVP into a credible pilot for 50 shelters in Lisbon and Porto, validate the predict-recommend-marketplace loop on European data, and reach financial break-even by month 12. The pilot launches with the prediction engine and recommendation flow as the critical path — these are the differentiating features and the cheapest to validate. The generative-AI features (description rewriting, Photo Studio, voice transcription, smart adopter search) are already built and available on the Pro tier, but their independent contribution to adoption speed will be measured through A/B testing before we claim credit for them.

The full AI Canvas sits in Appendix A.2. In summary: Prediction holds the verifiable per-listing probability distribution across five adoption-speed classes, with ground truth observable at 100 days. Action surfaces a color-coded priority badge on listing creation, generates a ranked recommendation list (add photo, lower fee, rewrite description with Gemini, enhance photo with FLUX, generate a shareable social-media sticker), routes performance views to shelter managers, and enables direct adopter-shelter chat — nothing executes without a human click. Judgement splits by user type: shelter managers decide which recommendations to act on; households decide whether to save, contact, or apply; the AdoptSense team sets the platform confidence threshold; longer-term match quality is judged post-adoption through Likert surveys. Outcomes are layered (model, listing, shelter, platform welfare), and the Feedback loop runs continuously: per-case ground truth at 100 days feeds quarterly regional retraining, while reinsertion and satisfaction signals throttle speed-optimizing features if welfare deteriorates.

6.2 Profitability Analysis

We apply the Expected Value Framework (EVF) to a binary recasting of the prediction (will this listing become long-stay (over 30 days) or short-stay) because the intervention is binary at the listing level. Full matrix and labelled assumptions sit in Appendix A.3.

On the shelter side, a long-stay listing costs approximately €15 per animal per day in blended boarding cost (Assumption A1 — conservative lower bound from shelter interviews; UK commercial kennels run £15–£26/day for reference). A successful intervention shortens stay by 14 days on average (Assumption A2 — primary recommendations like adding photos or adjusting fees are actionable within hours). Staff time per actioned recommendation runs about 20 minutes at €24/hour fully loaded, or €8 (Assumption A3). Expected benefit per true positive is $14 \times €15 - €8 \approx €200$. False negatives are assigned zero incremental cost relative to the status quo — a comparison artefact, not a claim that missed long-stays are costless in welfare terms (Assumption A4).

Model recall and precision on the binary recasting are working assumptions of 55 percent and 65 percent (Assumption A5 - they will be replaced with measured pilot numbers by month 6, and listing prioritization pauses automatically if precision falls below 50 percent).

At launch scale of ~10,000 platform listings across the 50 pilot shelters with 35 percent long-stay prevalence (~3,500 at-risk animals), the model flags ~1,925 true positives, generating gross operational benefit to shelters of approximately €385,000 against a false-positive cost of approximately €8,000 — a benefit-to-cost ratio around 48:1.

On revenue, the year-one model is SaaS-led: 20 Pro subscriptions at €49/month, ~30 featured-listing boosts at €10 each, modest platform advertising once shelter density makes ad inventory attractive. Year-one revenue is approximately €1,780/month against €2,200–3,000/month operating costs — closing near break-even, with the gap covered by Startup Portugal and EU EIC Accelerator grant applications. Unit economics scale sharply: at 500 Pro shelters, total AI inference cost is ~€186/month and fixed infrastructure ~€240/month against ~€24,500/month in revenue — a 98 percent gross margin (Appendix A.4).

6.3 Method Choice and System Design

The core adoption-speed prediction is best framed as classical supervised classification. The data is tabular and labelled, the prediction is narrow, and the 100-day verifiable outcome makes the classical machine-learning evaluation regime the natural fit.

The full AdoptSense system is a deliberate hybrid. Classical ML handles prediction. Generative AI (Google Gemini 2.5 Flash) handles language tasks: description rewriting, voice-memo transcription, text-to-speech narration, and parsing natural-language adopter queries into structured filters. Generative computer vision handles photo enhancement. Adopter-to-shelter chat is human-to-human — we deliberately do not auto-reply on shelters' behalf, given the emotional nature of adoption decisions.

On the autonomy spectrum (chat → copilot → agent), AdoptSense is firmly in copilot territory: the AI predicts, recommends, drafts, and enhances; the human reviews and clicks. Fully agentic features (automatic fee adjustments, automatic transfers, automatic adopter replies) were rejected because feedback loops in welfare contexts are slow, operational stakes affect animals and partner trust, and we lack the monitoring to detect quiet failures.

The system reads from the platform's own database and writes only logged predictions, recommendations, and human-confirmed actions. Controls include per-shelter confidence thresholds, per-recommendation human override, role-based access (households cannot see shelter dashboards or recommendations), an audit log, and the admin dashboard surfacing growth, listings, users, surveys, and a live event feed in real time. Evaluation runs at three cadences: continuous (model metrics, drift), weekly (recommendation acceptance, KPIs, satisfaction), quarterly (length-of-stay, reinsertion, regional retraining).

6.4 Risks and Mitigations

Five risks deserve operational specificity rather than general acknowledgement.

Geographic transfer error from Malaysian training data is the most immediate technical risk. Mitigation: from pilot month one, the data scientist collects Portuguese ground truth alongside live predictions; by month six we retrain on a Portugal-stratified sample. Predictions in low-data regions carry an explicit "low confidence - regional validation in progress" label until then. Shelter managers can dispute predictions in the admin console, and disputed cases feed the next retrain.

Bad-faith adoption and pet flipping is the biggest welfare risk. Mitigation: the admin dashboard logs every adoption and every re-listing matched on the platform's pet identifier and listing-photo fingerprint. Any user account showing more than two reinsertions in twelve months loses listing prioritization; platform-wide reinsertion exceeding 5 percent triggers automatic policy review and pauses all speed-optimizing features until the rate recovers.

Hallucinated AI descriptions overstating a pet's traits create both welfare and legal exposure. Mitigation: every AI-rewritten description requires explicit shelter approval before going live, the original raw description stays accessible, AI-generated content carries a visible badge, and repeat-offender shelter accounts face progressive restrictions through the admin console.

Platform abuse and scammers: mitigation includes mandatory verification for shelter managers (cross-checked against municipal directories), progressive verification for households, photo-metadata checks, a one-click report flow on every listing, and a 24-hour admin-team review commitment on reported accounts.

Welfare-versus-speed conflict: reinsertion and Likert KPIs operate as automatic gating signals; deterioration of more than 10 percent quarter-on-quarter pauses listing-prioritization and featured-listing features platform-wide until metrics recover.

6.5 Governance, Fairness, and Explainability

Governance. AdoptSense is a non-profit, which simplifies the governance frame. Decisions are governed by the welfare mission, not by shareholder returns. The data scientist owns model changes and retraining, the CEO owns commercial and pricing decisions, and welfare-policy decisions (network-wide pauses on prioritization, etc.) are documented and reviewable. The regulatory frame is the European Union AI Act, where AdoptSense's predictions and AI-generated content sit in the limited-risk category, carrying transparency obligations. We meet these by labelling every element created with AI with a visible disclosure badge, publishing a model card disclosing accuracy, training data origin, and known biases, and maintaining a Right-to-Erasure protocol. General Data Protection Regulation (GDPR) compliance for commercial launch requires Data Processing Agreements with Google and Hugging Face; these are scoped into the year-one budget.

Fairness. Our concern is not about legally protected human attributes but about which animals get algorithmic visibility - the risk that the model under-predicts adoption speed for certain breeds, ages, colours, or shelter regions, pushing those animals lower in priority and lengthening their actual stays.

Explainability. Every prediction shown to a shelter manager is paired with a plain-language explanation ("the main reason this listing scored 'High priority' is that it has only one photo and a fee above the regional average"). Where managers want a localized second view, Local Interpretable Model-agnostic

Explanations (LIME) are available on demand. Output is always paired with the natural-language recommendation badge rather than a raw probability. Disputed predictions enter the next quarterly retrain.

6.6 Call to Action

We request CEO approval to move AdoptSense from MVP to a 12-month Portugal pilot with a year-one budget of approximately €30,000: €18,000 for one part-time hire combining data-science and community-management responsibilities, €4,000 for infrastructure migration (PostgreSQL, cloud photo storage, FastAPI backend), €3,000 for commercial AI licences (FLUX Pro tier) and Data Processing Agreements, €3,000 for shelter-onboarding events and household marketing in Lisbon and Porto, €2,000 reserve. Funding combines NGO micro-grants, Startup Portugal, the EU EIC Accelerator pre-application, early Pro subscription revenue, and continued GoFundMe support. Approval includes the authority to negotiate pilot agreements with Lisbon and Porto municipal shelter networks, the mandate to publish the model card and governance policy, and explicit welfare-KPI gating at month six. We recommend proceeding.

Appendix

A.1 Peer Feedback Response - Adopted and Not Adopted

Peer review of the first draft surfaced themes across five reviewers. The summary below records which suggestions were adopted and which were not, with reasons.

Adopted.

Business model clarity: Section 4 and Appendix A.4 now name the Pro subscription at €49/month as the primary year-one revenue driver and map the four-phase financial trajectory (validate today; cover costs year 1; scale year 2–3).

Audience and tone: model jargon replaced with plain-language framing (e.g. XGBoost as "gradient-boosted decision tree", macro F1 as "balanced accuracy measure"; LLM, GDPR, SHAP, LIME spelled out on first use), technical specifics moved to the appendix, Section 4 reframed around market and financials.

Scope and prioritization: Section 6.1 identifies prediction and recommendations as critical-path, with generative-AI features measured as a deliberate second wave via A/B testing.

Unjustified assumptions: Appendix A.3 now contains a labelled assumption table (A1–A5) flagged in-text.

Governance and risks: Section 6.4 names who decides, what thresholds trigger action, on what timeline; Section 6.5 completes governance with EU AI Act framing, model cards, SHAP-based explanations, and class-imbalance discussion.

Other: Lessons Learned moved to dedicated Section 5; tenets rewritten in mission-grounded language; non-profit framing made explicit; macro F1 0.35 framed honestly as a limitation; GitHub link placed at top; CTA carries concrete €30,000 budget.

Not adopted.

The suggestion to merge "Introduction" and "Relevant Data" into a single packed opening section. We kept them separated because the Amazon template treats context-setting and supporting evidence as distinct beats, and a fully merged opening risked burying either the problem statement or the market data. The two sections now flow more tightly but remain structurally separate.

The recommendation to name first-pilot shelters and detail per-shelter onboarding inside the memo body. Naming concrete partnerships is not the objective of a strategy memo in our opinion. We have provide a multi-phase start recommendation without picking out concrete shelters, but rather regions.

A literal "+500 percent qualified inquiries" target as a headline goal. Rather than just defending the figure as suggested, we went further and demoted it from a target to upside language in Section 2, because it was generated on a very small MVP sample and would over-promise to a CEO audience. It can be validated later on.

A.2 AI Canvas — AdoptSense (Two-Sided Platform, Copilot Scope)

Business problem: minimise time-to-adoption and maximise long-term match quality across a two-sided pet adoption marketplace.

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	Prediction	**Action**	**Judgment**	**Outcomes**
	Will this listing become long-stay (>30 days)? Verifiable at 100 days. Full model: probability distribution over 5 adoption-speed classes.	Surface prediction on listing creation; surface ranked recommendations (add photo, adjust fee, enhance photo with FLUX, rewrite description with Gemini, create sticker); shelter-household chat. **No action executes without human approval.**	Shelter managers decide which recommendations to act on. Households decide whether to save, contact, or apply; they do not receive operational recommendations. **Platform team sets global confidence threshold.** Adopters and shelters judge match quality post-adoption via Likert surveys.	**Model: macro F1, per-class recall, calibration drift, AUC Speed 4.** Business: average length-of-stay, qualified inquiry-to-adoption conversion, reinsertion rate, Likert satisfaction.
Data	Platform listings, accounts, watchlists, messages, uploaded images, PetFinder.my warm start, regional partition labels	Listing CMS, recommendation templates, listing detail pages, chat, photo-enhancement endpoint, partner ad inventory	Per-listing outcomes, satisfaction surveys, reinsertion records	Prediction logs, action logs, outcome labels, satisfaction logs
Human	Data scientist	Shelter operators; households only browse, save, and chat	Shelter managers, adopters, household decision-makers	Shelter managers, data scientist
Other	Cloud ML platform	Listing platform, LLM API, image model	Audit and reporting flow	BI dashboard, quarterly retrain
Feedback	Per-case ground truth at 100 days → quarterly regional retrain. A/B-test recommendation logic. Continuous drift monitoring. Likert surveys and reinsertion rate close the welfare loop.			

Workflow change: shelter staff move from reactive monthly long-stayer review to proactive day-zero listing optimisation. Household adoption moves from fragmented browsing across classified sites to a guided, measurable demand channel connected directly to shelter managers.

A.3 EVF Matrix and Labelled Assumptions

Expected Value Framework benefit/cost matrix (base scenario: status quo, no AdoptSense intervention):

Actual → / ↓ Predicted	**Actual: Long-stay** *Guest stayed long-term*	**Actual: Short-stay** *Guest stayed short-term*
Predicted: Long-stay *(flag and recommend)*	**+€200** *True Positive (TP)*	**−€8** *False Positive (FP)*
Predicted: Short-stay *(no action)*	**€0** *False Negative (FN) — status-quo cost, not incremental*	**€0** *True Negative (TN)*

Labelled assumptions (referenced in Section 6.2):

ID	Assumption	Justification / Validation Plan
A1	Blended boarding cost ≈ €15 per animal per day at partner shelters.	Conservative lower bound from MVP shelter interviews; UK commercial kennels run £15–£26/day, Portuguese municipal facilities operate at scale with lower unit cost. Validated per-shelter in pilot month one.
A2	Each successful recommendation shortens stay by ≈ 14 days.	Mid-range estimate. Primary interventions (adding a photo, adjusting fee) are actionable within hours; the 14-day delta reflects the listing-quality gap observed in MVP analysis. A/B-tested with pilot shelters in months 2–6.
A3	Staff time per actioned recommendation ≈ 20 minutes at €24/hour fully loaded (€8).	Time-and-motion estimate from MVP user testing; the €24/hour figure is the EU average fully-loaded NGO rate for animal-welfare staff.
A4	False negatives carry zero incremental cost relative to the status quo.	Comparison artefact: the no-AdoptSense baseline already produces those long stays. This is not a claim that missed long-stays are costless in welfare terms; the welfare cost is captured separately through reinsertion and satisfaction KPIs.
A5	Binary-recasting recall ≈ 55%, precision ≈ 65%.	Derived from the 5-class measured macro recall of 0.34 and the 0.75 AUC on Speed 4 (no adoption). Replaced with measured pilot numbers by month 6. Listing prioritisation pauses automatically if precision falls below 50%.

A.4 Pricing, Revenue, and Token Economics

Pricing tiers:

Tier	Price	Included
Adopter (Households)	Free	Marketplace access, smart filters, AI match scores, watchlist, direct messaging.
Shelter Basic	Free	Unlimited listings, AI adoption-speed prediction, basic KPI tracking, inquiry management, 30-day free Pro trial.
Shelter Pro	€49 / month	Everything in Basic plus featured-listing placements, full KPI dashboard, listing-prioritisation tools, AI description improvement, token-based scaling, priority support.

Three-phase financial trajectory (from company presentation):

Phase	Revenue (monthly)	Cost & funding (monthly)
Phase 1 (today): Validate	€0 (volunteer team)	~€100/mo (hosting, Gemini API, domain); funded via GoFundMe, NGO micro-grants, early shelter contributions.
Phase 2 (Year 1): Cover costs	~€1,780/mo: 20 Pro × €49 + ~30 featured boosts × €10 + modest platform ads.	~€2,200–3,000/mo; gap covered by Startup Portugal and EU EIC Accelerator grants. Break-even target.
Phase 3 (Year 2–3): Scale	~€23,700/mo: 300 Pro × €49 + featured listings €4,000 + advertising €5,000.	Annual revenue ~€280,000; team salaries and operating costs fully covered; expansion to Spain/Italy.

Unit economics at scale (500 Pro shelters):

Cost item (at 500 Pro shelters)	Monthly cost
Studio photo enhancement (FLUX)	~€150 (€0.010/photo)
Text-to-speech narration	~€12 (€0.001/play)
Description rewrite	~€18 (€0.002/listing)
Smart filter query parsing	~€6 (€0.0004/query)
Google Gemini 2.5 Flash + HuggingFace FLUX.1-Kontext	~€186/mo (total inference across all AI features)
Fixed infrastructure (Cloud Run + DB + storage + CI/CD)	~€240/mo
Total operating cost at 500 shelters	**~€426/mo**
Revenue (500 Pro × €49)	**€24,500/mo**
Gross margin	**~98.3%**

A.5 MVP Model Performance Reference

Metric	Value	What does it mean?
Algorithm	XGBoost multi-class classifier (multi:softmax, 5 classes, 300 estimators)	The type of AI model used. It learns from thousands of past adoption listings and uses that experience to predict how quickly a new animal is likely to be adopted.
Feature count	31 (27 tabular + 4 VADER sentiment) in deployed pipeline	The number of pieces of information the model looks at for each animal: things like age, breed,

		whether there's a photo, and the tone of the listing description.
Validation accuracy (5-class)	0.3991 (vs. 0.20 random baseline)	correct predictions / total predictions on held-out data. A random guess across 5 classes scores 0.20; 0.40 means the model is ~2× better than chance.
Macro F1	0.3461	mean F1 across all 5 classes (unweighted). F1 = 2 × (precision × recall) / (precision + recall). Macro averaging treats every class equally regardless of size: penalises poor performance on rare adoption speeds.
Weighted F1	0.3877	F1 per class, weighted by class frequency, then summed. Higher than Macro F1 here because the model performs better on more common classes; gives a realistic picture of overall day-to-day accuracy.
AUC, Speed 4 (no adoption at 100 days)	~0.75	Out of every 4 animals flagged as high-risk, the model correctly identifies 3 of them.
Training corpus	PetFinder.my Kaggle dataset, ~15,000 Malaysian listings	The model learned from 15,000 real pet listings. Since these come from Malaysia, the patterns it learned may not perfectly match European shelters
Top feature importances	has_photo (0.129), Sterilized (0.050), age_bin (0.042), Age (0.039), photo_bin (0.039)	The factors that most influence the model's predictions. Having a photo is by far the strongest signal, followed by whether the animal is sterilised and its age. These mirror what shelter staff often observe in practice.
Label framing	Raw classes 0–4 surfaced as recommendations ("Top listing", "List now", "Optimize listing", "Needs attention", "High priority") with confidence; never as time promises.	The model's output is a number (0–4), but staff see it as a plain-language action — such as "List now" or "Optimise listing" — so no technical knowledge is needed to act on the result.

A.6 References

- Agrawal, Gans and Goldfarb (2018, 2022) for the AI Canvas and prediction-judgement-action framework.
- European Union AI Act (Regulation (EU) 2024/1689), transparency obligations for limited-risk AI systems.
- FEDIAF (European Pet Food Industry Federation) household pet ownership figures, 2024 (€6.74 billion European pet market; 139 million households).
- German Animal Welfare Federation (Deutscher Tierschutzbund) shelter capacity survey, 2024 (cited via Euronews, 2025).
- Lundberg, S. and Lee, S.-I. (2017), "A Unified Approach to Interpreting Model Predictions" — SHapley Additive exPlanations (SHAP).
- PetFinder.my Adoption Prediction dataset, Kaggle. https://www.kaggle.com/c/petfinder-adoption-prediction
- Portuguese municipal shelter intake figures, 2024 (42,718 animals entered shelters; 10,171 not adopted; 2,421 euthanised; cited in AdoptSense company presentation).
- Ribeiro, M. T., Singh, S., and Guestrin, C. (2016), "'Why Should I Trust You?' Explaining the Predictions of Any Classifier" — Local Interpretable Model-agnostic Explanations (LIME).
- Spanish dog abandonment figures, 2024 (~300,000 dogs cited in Euronews, 2025).

A.7 Use of AI in the Preparation of This Memo and the MVP

This memo and the AdoptSense MVP itself were both built with AI assistance, structured deliberately to keep design ownership with the team.

For the memo, the main tool was Anthropic's Claude (Claude Opus 4.7, chat-based interface). The development strategy was design-first: Claude proposed a memo structure reconciling the Amazon 6-page template with the assignment's required content sections and AdoptSense's two-sided business model; the structure was reviewed and adjusted before drafting began. Specific iterations rebuilt the profitability section to align with MVP findings, sourced third-party statistics rather than inventing numbers, framed assumptions explicitly in the appendix, and trimmed prose to the 3,000-word limit. After every draft, the team reviewed, commented, and revised manually before the next iteration. The the non-profit business model, the MVP findings, the decision to request a pilot is the team's; the AI's contribution was structural, editorial, and a search assist for sourced market statistics.

For the MVP, two AI assistants were split by role. Claude Code (the VS Code extension, Anthropic Opus 4.7) handled agentic work — repository reads and writes, multi-step refactors, file edits — gated through a "plan-then-approve" pattern where every change required a written plan from Claude before any code was committed. Claude.ai (web) ran in parallel for feasibility discussion and presentation drafting. ChatGPT (OpenAI GPT-5.5 Thinking) handled brainstorming and experimental design outside the integrated development environment, and served as a second-opinion code review channel that prevented anchoring on Claude's earlier suggestions. Version control through GitHub tracked every change and enabled parallel collaboration.

Three working modes kept design ownership with the team. Design-first: benchmark architecture proposed by Claude, iterated by the team before any code was written. Debugging-led: feature engineering and modelling attempted by hand first, with the AI escalated only on failure. Benchmark-guided: after each training run, the team fed profiling output back to Claude with a single question: "where is the next bottleneck?" letting performance data drive the conversation rather than open-ended prompts.

TeamShape was not used for two reasons. First, AdoptSense is a real project built together with an external team, and logging individual contributions inside an assessment platform would mean taking credit for other students' work, which is something this team isn't comfortable with. Second, free-riding simply wasn't an issue; the team communicated and collaborated well throughout, so TeamShape would have been solving for a problem this team didn't have.

What worked: the agentic-plus-chat split moved ideas from sketch to running code in minutes, the plan-then-approve gate gave agentic speed without losing design ownership, and the benchmark-guided loop produced concrete next steps rather than generic advice. What did not: open-ended prompts produced over-engineered solutions until we standardized on "the smallest change that achieves X", and long Claude Code sessions lost track of the database schema until we standardized on re-pasting schema and latest results at the top of each new task. The net effect was faster implementation that freed time for deeper assignment thinking, while the approval gate kept every committed file authored by the team.

A.8 Technical details

The predictive core is an XGBoost classifier - a gradient-boosted decision tree model, chosen for its performance on structured tabular data - trained on roughly 15,000 PetFinder.my listings and serving real-time predictions across 31 input features. Generative AI runs end-to-end alongside it: Google Gemini handles description rewriting, voice-memo transcription, text-to-speech narration, and natural-language adopter search, while FLUX.1-Kontext via Hugging Face powers the in-app Photo Studio, removing backgrounds and enhancing listing images automatically. The marketplace ties it together — persistent listings, watchlists, household-shelter chat, an interactive shelter map, engagement surveys, and a sixteen-metric KPI dashboard for shelter managers. The MVP runs on Streamlit and SQLite, which were the right tools for rapid iteration but are not production-ready. Migrating the UI layer to FastAPI, moving the database to PostgreSQL with cloud photo storage, and executing Data Processing Agreements with Google and Hugging Face are non-negotiable pre-launch tasks, already scoped into the year-one budget.

A.9 AdoptSense: Project Documentation & Quick Start Guide

Repository Architecture
The project is organized into two primary layers: a backend ML pipeline and a frontend marketplace application, connected through a centralized configuration system.

Backend Pipeline & Model Layer:
The machine learning backbone resides in the src directory and handles data preparation, feature engineering, and predictive modeling:

- config.py — Central configuration container that resolves all filesystem paths relative to the project root, ensuring the system works seamlessly regardless of working directory
- features_tabular.py — Extracts and engineers tabular features (breed, age, health status, vaccination records, adoption fees, pet characteristics)
- features_sentiment.py — Processes textual descriptions and sentiment analysis to derive behavioral and contextual features
- model/ — Contains the trained XGBoost multi-class classifier that predicts adoption speed into five classes (0–4), translating to actionable recommendations for shelter managers
- petadoption_run.ipynb — Complete exploratory data analysis, feature extraction, model training, and validation pipeline

Frontend Marketplace Layer:
The user-facing application is built on Streamlit and handles authentication, marketplace operations, and analytics:

- app.py — Main entry point providing navigation between all marketplace views
- utils/auth.py — Manages user authentication and role-based access control for three user personas (Shelter Manager, Private Household, Admin)
- utils/db.py — SQLite database abstraction layer managing listings, user accounts, conversations, watchlists, and adoption tracking
- utils/matching_platform_ui.py — Core marketplace views including pet browsing, detailed listing pages, listing creation/editing, KPI dashboards, AI-powered smart filtering, and direct messaging between shelters and households
- utils/admin_ui.py — Administrative dashboard for system oversight and governance
- components/ — Reusable UI components including the persistent navigation bar, pet listing cards, and authentication overlays
- styles.py — Global styling and CSS injection for consistent visual branding

Quick-Start Guide can be found in the README.md